



BRITISH INTERNATIONAL
SCHOOL OF SULAYMANIYAH

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Information & Communication Technology



OBJECTIVES

- ❖ **Understand** the characteristics and constraints of communication via emails.
- ❖ **Learn** about corporate email policies, security, and the necessity of netiquette.
- ❖ **Differentiate** between the Internet and the World Wide Web (WWW).
- ❖ **Explore** the uses and advantages of intranets, extranets, blogs, wikis, and forums.
- ❖ **Understand** the roles of ISPs, search engines, URLs, and internet protocols.
- ❖ **Evaluate** internet information and understand the debate around internet risks.

- ❖ Email is one of the most common communication methods used globally.
- ❖ Specific rules are required to ensure message security and privacy.
- ❖ Guidelines prevent the transmission of socially unacceptable content.
- ❖ Both individuals and corporate companies must adhere to these constraints.
- ❖ Many countries have enacted legislation to govern appropriate email system use.

- ❖ National laws exist specifically to protect people against the misuse of emails.
- ❖ Senders are often legally required to obtain 'opt-in' permissions from users.
- ❖ Companies must provide a clear and simple 'opt-out' option for recipients.
- ❖ A valid postal address must accompany corporate or organizational emails.
- ❖ Organizations must clearly communicate their privacy policies to subscribers.
- ❖ Harvesting email addresses without user consent is widely prohibited.

- ❖ Users must adhere to an acceptable code of practice when drafting messages.
- ❖ The transmission of obscene images or abusive text is strictly forbidden.
- ❖ Profane, inflammatory, defamatory, or coercive language is unacceptable.
- ❖ Racist, exploitative, or violent messages are strictly prohibited.
- ❖ The dissemination of illegal materials or messages is not allowed.
- ❖ Simply assuming recipients can delete offensive emails is not an adequate defense.

Employer Email Guidelines

- ❖ Companies must publish clear guidelines regarding electronic communication.
- ❖ Company email systems are primarily designated for authorized business use.
- ❖ The acceptable style and professional tone of emails should be standardized.
- ❖ Employees must only use designated, password-protected corporate accounts.
- ❖ Clear rules regarding data storage and information confidentiality must exist.
- ❖ Ongoing training ensures staff continuously follow company email policies.

- ❖ Emails and any attached files are fully subject to copyright laws.
- ❖ Printing or forwarding an email may breach copyright without sender permission.
- ❖ Recipients must verify confidentiality status before forwarding any messages.
- ❖ Corporate emails often include specific legal copyright and disclaimer statements.
- ❖ Forwarded emails may trigger ISP spam filters by masking the true sender.
- ❖ ISPs may blacklist domains that frequently forward large email volumes.

- ❖ Strong passwords must combine letters, numbers, and specialized symbols.
- ❖ System passwords should be changed on a regular, scheduled basis.
- ❖ Spam filters help redirect suspicious emails directly to a 'junk' folder.
- ❖ Anti-virus software must run constantly to block threats from malicious sources.
- ❖ Emails remain vulnerable to unauthorized reading if left completely unprotected.

- ❖ Passive attacks involve the unauthorized release of private email materials.
- ❖ Active attacks actively modify messages without the sender's knowledge or consent.
- ❖ Denial of service (DoS) overloads systems by sending thousands of junk emails.
- ❖ Active attacks frequently deploy viruses directly to the recipient's computer.
- ❖ Phishing attacks use deceptive emails to steal sensitive user data and passwords.

- ❖ Netiquette is a modern blend of the words 'interNET' and 'etIQUETTE'.
- ❖ It represents a standard code of common courtesy for online communication.
- ❖ Writers must respect other users' differing views and online opinions.
- ❖ Readers cannot rely on facial expressions to help interpret a message's tone.
- ❖ Humor and sarcasm can easily be misunderstood in written electronic formats.

- ❖ Do not be abusive, threaten people, or engage in personal attacks.
- ❖ Be clear and succinct in your messaging; avoid unnecessary waffle.
- ❖ Always check spelling and grammar to maintain a professional impression.
- ❖ Respect privacy and do not publish information that might embarrass others.
- ❖ Avoid using CAPITAL LETTERS, as this is universally perceived as shouting.
- ❖ Do not plagiarise; always appropriately acknowledge quotes and external sources.

- ❖ Groups allow sending single messages efficiently to multiple targeted addresses.
- ❖ Companies use groups for marketing based on user demographics or hobbies.
- ❖ They are highly efficient for organizing meetings and corporate video conferences.
- ❖ Groups ensure no intended recipient is accidentally forgotten from an invite.
- ❖ However, malicious spammers can exploit groups by raiding internal address books.
- ❖ This can result in massive spam distribution from a single compromised account.

- ❖ Carbon copy (Cc) makes email addresses fully visible to all other recipients.
- ❖ Cc is generally used simply to keep interested third parties informed.
- ❖ Blind carbon copy (Bcc) hides email addresses from all other recipients.
- ❖ Bcc acts as a vital privacy and security measure when emailing large groups.
- ❖ It protects individual user privacy by concealing personal contact details.
- ❖ Senders should place their own address in the 'To' field when utilizing Bcc.

- ❖ Attachments from unknown senders should be treated with extreme caution.
- ❖ Malicious attachments can easily infect local computers with harmful viruses.
- ❖ Executable files (.exe, .bat, .msi) carry the absolute highest security risks.
- ❖ Internet Service Providers generally impose limits on allowable file attachment sizes.
- ❖ Large files can be safely compressed into zip files for easier transmission.
- ❖ Cloud web services can host large files securely and provide a download link.

- ❖ Spam refers to any unsolicited bulk email sent over the internet network.
- ❖ It is typically distributed to a massive number of recipients simultaneously.
- ❖ Spam ranges from mildly annoying mass advertising to highly dangerous scams.
- ❖ It frequently contains hidden viruses or malicious links to phishing websites.
- ❖ Spammers utilize automated software called 'spambots' to distribute links rapidly.
- ❖ Mobile phones can also receive highly annoying spam via text messages (m-spam).

- ❖ Sorting through and deleting spam wastes a significant amount of user time.
- ❖ It is universally perceived as a major nuisance by the general public.
- ❖ High volumes of spam unnecessarily consume valuable internet network bandwidth.
- ❖ This massive bandwidth consumption measurably slows down overall internet speeds.
- ❖ It frequently clogs up user inboxes, easily burying legitimate communication.
- ❖ Mobile phone users may incur unfair financial charges for receiving spam SMS.

S25_QP11_Q12

When sending emails users should follow netiquette.

(a) **Explain** what is meant by netiquette.

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Question	Answer	Marks
12(a)	Two from: The correct / acceptable way of communicating over the internet Internet etiquette Respects other user's views Displays common courtesy when sending messages	2

M25_QP12_Q5

The cloud can be used for computer gaming. One major issue is the protection of personal data.

- (c) One player wants to send the list of players and their scores to a friend as an attachment.

State the precautions that need to be taken when sending and receiving the attachment.

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Question	Answer	Marks
5(c)	Three from: Install anti-virus software Keep the anti-virus software up to date Scan the attachment for viruses Remove all personal data from the document Encrypt / password protect the file if personal data is being sent Check that it is being sent to the correct email address Make sure the user is sending the correct attachment	3

- ❖ The Internet is a worldwide, physical collection of interconnected computer networks.
- ❖ The Internet relies heavily on transmission control protocols (TCP) and IP addresses.
- ❖ The World Wide Web (WWW) is a massive digital collection of multimedia pages.
- ❖ The WWW is essentially a service accessed using the underlying internet infrastructure.
- ❖ Web pages are fundamentally built using Hypertext Markup Language (HTML) documents.
- ❖ The WWW is accessed and navigated using web browsers and HTTP protocols.

- ❖ Intranets are internal computer networks built entirely on internet technology.
- ❖ They are specifically designed to share information within a single organization.
- ❖ Access is strictly confined to company members and explicitly authorized users.
- ❖ Intranets are robustly protected from the public internet by a network firewall.
- ❖ They require specific, authenticated user IDs and passwords for initial access.
- ❖ Information is usually stored on highly secured local organizational servers.

- ❖ They are exponentially safer because external hacking or viruses are minimized.
- ❖ Administrators can easily block internal access to inappropriate external websites.
- ❖ Company information can be strictly tailored specifically to internal audience needs.
- ❖ Sensitive corporate messages safely remain enclosed within the company network.
- ❖ Intranets typically offer much better, dedicated bandwidth than the public internet.
- ❖ Internal data transfer speeds are usually significantly faster and more reliable.

- ❖ Extranets function as secure extensions of a company's internal intranet framework.
- ❖ They allow strictly limited access to specific users outside the immediate organization.
- ❖ They share the exact same underlying technology and security advantages as intranets.
- ❖ External access is carefully controlled using robust password authentication measures.
- ❖ Trading partners can safely access necessary, commercially sensitive shared information.
- ❖ Virtual Private Networks (VPNs) can securely connect remote workers to these networks.

- ❖ Web logs (blogs) function as personal internet journals for individual writers.
- ❖ They are updated regularly and automatically displayed in reverse chronological order.
- ❖ Blogs cover diverse topics ranging from minor personal hobbies to major social issues.
- ❖ Bloggers can still be legally prosecuted for posting offensive or defamatory material.
- ❖ Microblogs feature very short, frequent posts common on social media networks.
- ❖ Business blogs (b-blogs) are effectively used by companies for internet promotion.

- ❖ Wikis are web applications allowing authorized users to create and edit pages.
- ❖ They operate using a very simple, accessible syntax known as wiki markup.
- ❖ Multiple distinct authors can collaboratively edit, delete, or modify shared content.
- ❖ The core software automatically keeps a detailed history of all document revisions.
- ❖ They allow massive, complex documents to be easily shared among many people.
- ❖ Web pages can be organically organized according to the authors' specific wishes.

- ❖ These sites aim to build online communities for users with shared social interests.
- ❖ Members are provided with free digital web space to build robust public profiles.
- ❖ Users can upload diverse content such as text messages, photos, and live videos.
- ❖ Instant messaging and live video chatting features are usually natively integrated.
- ❖ Members have critical tools to control who can access their private personal data.
- ❖ They greatly facilitate relationship building and rapid communication between friends.

- ❖ Forums act as online discussion groups focused tightly on specific core topics.
- ❖ Moderated forums feature active administrators who approve posts before publication.
- ❖ Moderators proactively filter out spam, rude content, or entirely off-topic messages.
- ❖ The open internet itself is largely an unmoderated, unpoliced digital forum.
- ❖ Users must heavily rely on voluntary cooperation and general network operators' rules.
- ❖ Breaching established forum protocols usually results in immediate post deletion or bans.

- ❖ ISPs are commercial enterprise companies providing users with internet access.
- ❖ Users typically pay a recurring monthly subscription fee for these connectivity services.
- ❖ ISPs set up localized accounts featuring unique user IDs and secure passwords.
- ❖ They provide all the necessary physical equipment and telecommunications lines.
- ❖ Fast connections are usually established via broadband connections or fibre optic cables.
- ❖ Some ISPs actively monitor user email volumes to effectively filter out massive spam.

- ❖ Browsers are specialized software used to display web pages on computer screens.
- ❖ They effectively translate raw HTML code into visible text, videos, and images.
- ❖ Browsers locally maintain a history of visited sites and store user bookmarks.
- ❖ They utilize built-in navigation tools to move backward and forward between pages.
- ❖ Hyperlinks enable users to instantly click and navigate between different web pages.
- ❖ Links are usually displayed simply as blue underlined text or clickable digital images.

- ❖ URLs act as specific, standardized text addresses used to access websites and files.
- ❖ They are physically typed directly into the web browser's designated address bar.
- ❖ A URL strictly begins with the communication protocol (such as http or https).
- ❖ The website address includes the main domain host, specific name, and type (.com).
- ❖ Specific country codes (.uk, .us) are sometimes appended to the domain ending.
- ❖ The file path and exact filename specify the exact web page being requested.

- ❖ Search engines help users locate specific information across millions of web pages.
- ❖ They operate using massive, proprietary indexed databases of crawled websites.
- ❖ Users type a specific search string or criteria directly into the main search box.
- ❖ The engine immediately returns a prioritized list of matching results known as 'hits'.
- ❖ Advanced search interface options help reliably reduce results to a manageable number.
- ❖ The actual sorting algorithms determining specific search rankings are complex and secret.

- ❖ Online information is incredibly vast and virtually limitless in overall digital scope.
- ❖ Web pages can be updated instantly, keeping information highly current and relevant.
- ❖ Search engines make locating highly specific data fast, intuitive, and user-friendly.
- ❖ Deep research can be conducted comfortably from home without physical library visits.
- ❖ The vast majority of baseline information on the internet is completely free to access.
- ❖ Rich multimedia elements like videos make learning significantly more engaging.

- ❖ The internet is mostly unregulated, meaning absolutely anyone can publish anything.
- ❖ Information may be heavily biased, factually incorrect, or lack peer review.
- ❖ Search engines carry the inherent risk of presenting inappropriate or distressing sites.
- ❖ Users can easily become highly distracted by games or social networking platforms.
- ❖ Inexperienced researchers may frequently suffer from severe 'information overload'.
- ❖ Easy digital 'copy and paste' functions create a massive temptation for plagiarism.

- ❖ Always deliberately attempt to verify the basic legitimacy of the author or source.
- ❖ Cross-reference claimed facts with other reliable websites to ensure correctness.
- ❖ Check basic grammar and spelling, as obvious errors often indicate unreliable sources.
- ❖ Look critically for evidence of heavy bias or advertising influencing the content.
- ❖ Check the specific publication date to ensure the referenced information is not outdated.
- ❖ Verify if bold arguments are supported by reliable external links or recognized experts.

Internet Protocols: HTTP & FTP

- ❖ Protocols are strictly agreed communication rules between sender and recipient devices.
- ❖ Hypertext Transfer Protocol (HTTP) securely transfers website data across the internet.
- ❖ Web browsers automatically default to HTTP if a specific protocol is entirely omitted.
- ❖ File Transfer Protocol (FTP) moves dedicated files between connected network computers.
- ❖ FTP operates very similarly to HTTP but is dedicated solely for file management.
- ❖ Both essential protocols ensure data is transferred efficiently, safely, and correctly.

- ❖ SSL is a critical protocol allowing highly secure, encrypted online data transmission.
- ❖ When actively combined with HTTP, it forms HTTPS for secure internet web browsing.
- ❖ HTTPS encrypts data so only the specific user and destination server can interpret it.
- ❖ Users can easily identify active SSL by a small padlock symbol in the browser bar.
- ❖ Digital SSL certificates are utilized to actively authenticate secure, registered websites.
- ❖ It is absolutely essential for online banking, shopping, and secure email exchanges.

- ❖ VoIP allows live audio and video chatting directly over active internet connections.
- ❖ It seamlessly converts analog voice signals into digital data packets for rapid transfer.
- ❖ It is frequently built into and integrated into instant messaging software platforms.
- ❖ Major social networking sites natively utilize VoIP to enable live digital video chats.
- ❖ It serves as a highly cost-effective alternative to utilizing traditional telephone networks.
- ❖ Stable, high-speed broadband internet is generally required for smooth VoIP communication.

- ❖ Active policing would reliably prevent illegal material from being posted publicly.
- ❖ It would effectively stop vulnerable groups and children from accessing harmful content.
- ❖ Strict controls could confidently prevent incorrect or dangerous information from spreading.
- ❖ It makes it significantly harder for internet novices to find highly inappropriate materials.
- ❖ Baseline regulation could ensure a consistently safer digital environment for all users.
- ❖ It could successfully limit the broad dissemination of criminal instructions or tutorials.

- ❖ Harmful material is frequently already widely available through alternative media sources.
- ❖ Policing the entire global internet would be prohibitively expensive to manage properly.
- ❖ Enforcing strict global rules is nearly impossible due to differing international laws.
- ❖ Broad policing directly infringes upon fundamental, democratic freedom of speech rights.
- ❖ Existing national laws can already be effectively used to prosecute illegal online behaviors.
- ❖ Determining what is legally 'offensive' is entirely subjective to different cultural groups.

M25_QP12_Q12

Parents are worried about their children's use of the internet.

- (a) **Explain** how parents could monitor the use of the internet by their children.

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Question	Answer	Marks
12(a)	Three from: Monitor browsing / search history Only allow children to work under parent observation Install website monitoring software Record the activities the children are doing online / websites visited	3